

DATABASES:

PostgreSQL w/ pgvector:

Pros:

- Stores all information in one DB (Discord users, jobs, resumes, etc.)
- Has native JSONB, since JD is a JSON should work seamlessly
- Easy to connect from python

Cons:

- Requires more setup than ChromaDB
- At an extremely high vector-search scale, a dedicated vector database can be easier to tune (might not apply for our use-case).

Pinecone:

Pros:

- Minimal infrastructure work.
- Straightforward Python API.
- Handles scaling, availability, and index management for you.

Cons:

- Paid Service
- Less control than a hosted version
- We still need another DB for users and other information

ChromaDB:

Pros:

- Very easy python setup
- Can run locally for testing without a separate server
- Stores documents, embeddings, and metadata
- Good for prototypes/proof-of-concept

Cons:

- Mainly meant for testing or development
- Chroma recommends a server-backend deployment for production
- Not the best choice for the bot's relational data

Embedding Models:

Paid Models:

- 1. OpenAI text-embedding-3-small**
 - a. \$0.02 per 1M tokens
 - b. Pros: Inexpensive + simple API + fast
 - c. Cons: Resume never leaves the server so needs internet
- 2. OpenAI text-embedding-3-large**
 - a. \$0.13 per 1M tokens
 - b. Pros: Better retrieval quality, multilingual, flexible dimensions
 - c. Cons: 6.5x more expensive than the small model
- 3. Voyage voyage-4-lite**
 - a. \$0.02 per 1M tokens after free allowance
 - b. Pros: Designed for retrieval + multilingual and has long context
 - c. Cons: Needs external API + separate account

Free/Local Models:

- 1. BAAI/bge-small-en-v1.5**
 - a. Small, Fast good for english retrieval
- 2. nomic-ai/nomic-embed-text-v1.5**
 - a. Longer context and strong retrieval
- 3. google/embeddinggemma-300m**
 - a. Larger and slower than BGE small, but stronger retrieval